2D Ideal MHD Solver on the Orszag-Tang Vortex

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Outline

1. Introduction and Problem Motivation
2. Mathematical Model
3. Numerical Model
4. Validation and Results
5. Discussion of Results
6. Conclusions

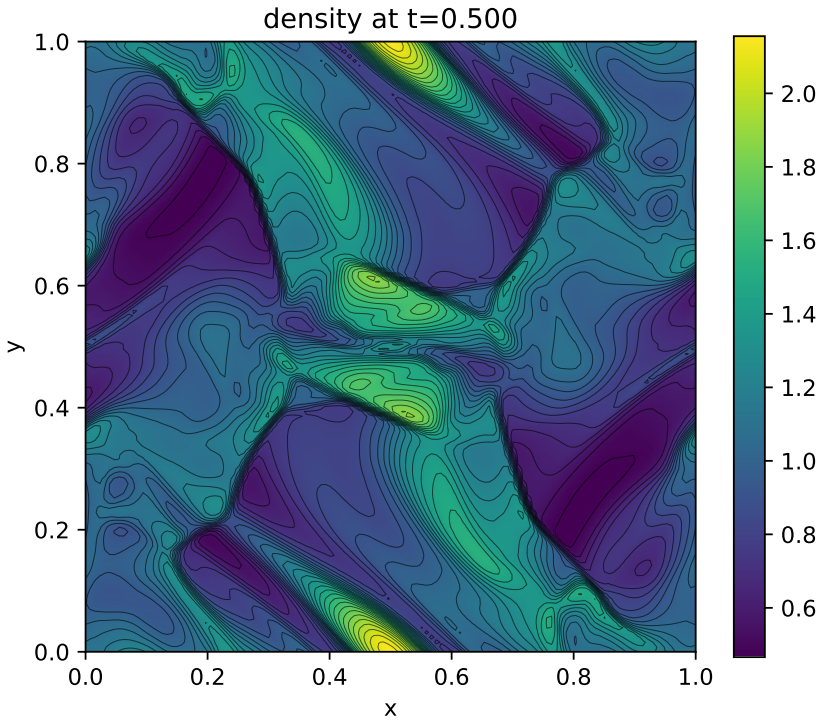
Why MHD

- Conducting fluid in motion = distributed current; *Lorentz force* $\mathbf{J} \times \mathbf{B}$ acts on every cell [1, 2].
- *Frozen-in flux*: in the ideal limit, field lines move with the fluid.
- Applies wherever plasma is collisional and magnetic Reynolds is large: solar wind, tokamaks, astrophysical jets, magnetospheres.
- New physics vs Euler:
  - *Alfven* waves—transverse, like waves on a tense string.
  - *Fast/slow magnetosonic*—sound waves modified by magnetic pressure.
  - Maxwell stresses $-B_i B_j$ in the momentum flux.
  - No-monopoles constraint $\nabla \cdot B = 0$.

Project goal: 2D ideal MHD finite-volume solver in Python, validated on Orszag-Tang.

Why Orszag-Tang

- Canonical 2D MHD benchmark.
- All-sinusoidal ICs on $[0, 1]^2$, periodic on every side.
- No geometry, no density jump, no outflow BCs.
- If the result is wrong, it is the solver, not the setup.
- Reference: [3, 4].



density $\rho(x, y,\ t = 0.5)$

Ideal MHD in conservative form

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} + \frac{\partial G(U)}{\partial y} = 0, \qquad U = (\rho,\ \rho u,\ \rho v,\ \rho w,\ B_x,\ B_y,\ B_z,\ E)^T.$$

Closure (ideal gas, $\gamma = 5/3$):

$$E = \frac{p}{\gamma - 1} + \tfrac{1}{2}\rho|\mathbf{v}|^2 + \tfrac{1}{2}|\mathbf{B}|^2, \qquad p_T = p + \tfrac{1}{2}|\mathbf{B}|^2.$$

- ρ: mass density. $\mathbf{v} = (u, v, w)$: velocity. $\mathbf{B} = (B_x, B_y, B_z)$: magnetic field.
- E: total energy density (internal + kinetic + magnetic).
- p: gas pressure. $\frac{1}{2}|\mathbf{B}|^2$: *magnetic pressure*. p_T: total (gas + magnetic).
- All 3 velocity and field components carried even in 2D so out-of-plane Alfven waves resolve correctly [5].
- Seven-wave fan: 2 fast, 2 Alfven, 2 slow, 1 entropy [6, 7].

x-direction flux $F(U)$

Row	$F(U)$ component	Physical meaning
1	ρu	mass flux (Euler)
2	$\rho u^2 + p_T - B_x^2$	x-momentum: total pressure p_T, tension $-B_x^2$
3	$\rho uv - B_x B_y$	Maxwell shear stress
4	$\rho uw - B_x B_z$	Maxwell shear stress
5	0	B_x self-flux $= 0$ in ideal MHD; GLM fills it
6	$uB_y - vB_x$	induction-equation flux for B_y
7	$uB_z - wB_x$	induction-equation flux for B_z
8	$(E + p_T)\,u - B_x(\mathbf{v}\cdot\mathbf{B})$	enthalpy carry + Poynting flux

- Rows 6–7 are components of the motional EMF $\mathbf{v}\times\mathbf{B}$ that drag field lines with the fluid (frozen-in flux).
- Row 8 carries Euler enthalpy $(E + p_T)u$ *plus* the Poynting flux $-B_x(\mathbf{v}\cdot\mathbf{B})$, which transports magnetic energy with the fluid.

y-direction flux $G(U)$

Row	$G(U)$ component	Physical meaning
1	ρv	mass flux (Euler)
2	$\rho v u - B_y B_x$	Maxwell shear stress
3	$\rho v^2 + p_T - B_y^2$	y-momentum: total pressure p_T, tension $-B_y^2$
4	$\rho v w - B_y B_z$	Maxwell shear stress
5	$v B_x - u B_y$	induction-equation flux for B_x
6	0	B_y self-flux $= 0$ in ideal MHD; GLM fills it
7	$v B_z - w B_y$	induction-equation flux for B_z
8	$(E + p_T)\, v - B_y(\mathbf{v}\cdot\mathbf{B})$	enthalpy carry + Poynting flux

- Same structure as $F(U)$ with (u, B_x) and (v, B_y) swapped.
- Implementation reuses the x-direction Riemann solver after the coordinate swap; halves the code surface.

The divergence constraint

- $\nabla \cdot B = 0$ is Gauss's law: no monopoles $\Rightarrow$ no field-line sources.
- Exact MHD preserves this; discrete updates leave a small divergence at every cell [8, 5].
- Why it matters: rewrite the Lorentz force as

$$(\nabla \times B) \times B = (B \cdot \nabla)B - \tfrac{1}{2}\nabla|B|^2 - (\nabla \cdot B)B.$$

  The last term vanishes analytically; once $\nabla \cdot B$ drifts, it acts as a spurious force parallel to B and the run blows up.
- Fix here: **GLM** cleaning [9]—one scalar ψ, drops into unstaggered FVM. (CT, Powell are alternatives.)

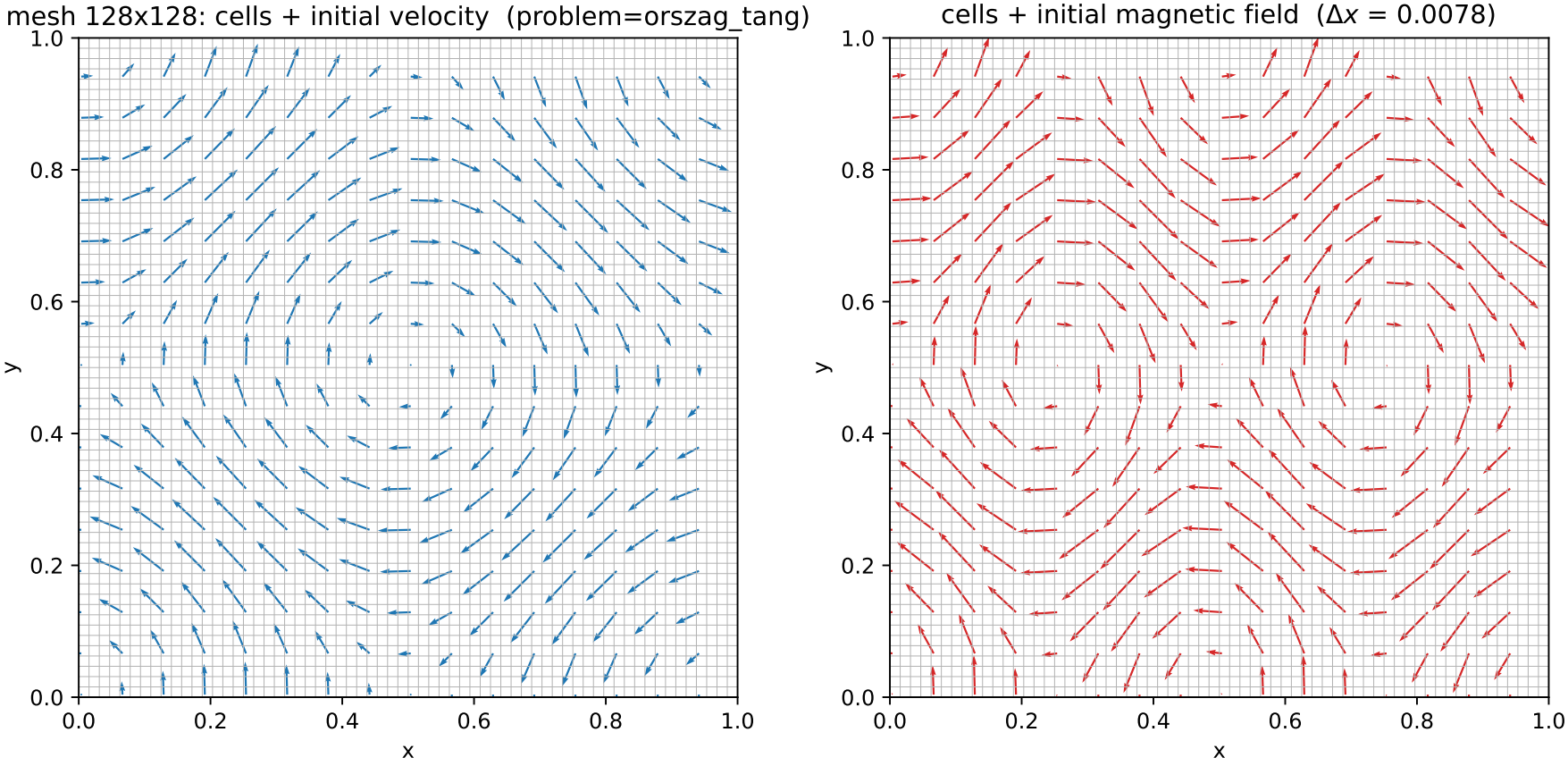
Computational mesh

- Uniform 128×128 Cartesian grid, $\Delta x = \Delta y = 1/128$.
- Two ghost layers per side (periodic fills for Orszag-Tang).
- ICs (Londrillo & Del Zanna [3]): $\rho = 1$, $p = 1/\gamma$, $B_0 = 1/\gamma$, and

$$\mathbf{v}(x, y, 0) = (-\sin 2\pi y,\ \sin 2\pi x,\ 0),$$
$$\mathbf{B}(x, y, 0) = B_0\,(-\sin 2\pi y,\ \sin 4\pi x,\ 0).$$

- Left panel: $\mathbf{v}$ overlay. Right panel: $\mathbf{B}$ overlay.

Computational mesh



Scheme at a glance

- Cell-centered Cartesian grid, two ghost layers [10, 11].
- MUSCL on primitives with minmod (MC, van Leer optional) [12].
- HLLD Riemann solver [7]; reduces to HLLC for $B = 0$.
- GLM ψ/B_x coupling baked into the Riemann solver; parabolic damping per RK stage [9].
- SSP-RK2 time integration [13], CFL $= 0.3$.

HLLD wave pattern

- Five-wave approximation to the full seven-wave MHD fan [7].
- Bounding speeds S_L, S_R from Davis-Einfeldt; entropy speed S_M from jump conditions.
- Two starred + two double-starred states across the rotational Alfven discontinuities.
- Positivity-preserving for physical inputs.

<table>
<tr><td>L</td><td>L^*</td><td>L^{**}</td><td>R^{**}</td><td>R^*</td><td>R</td></tr>
<tr><td colspan="2">S_L</td><td colspan="2">S_L^*</td><td>S_M</td><td>S_R</td></tr>
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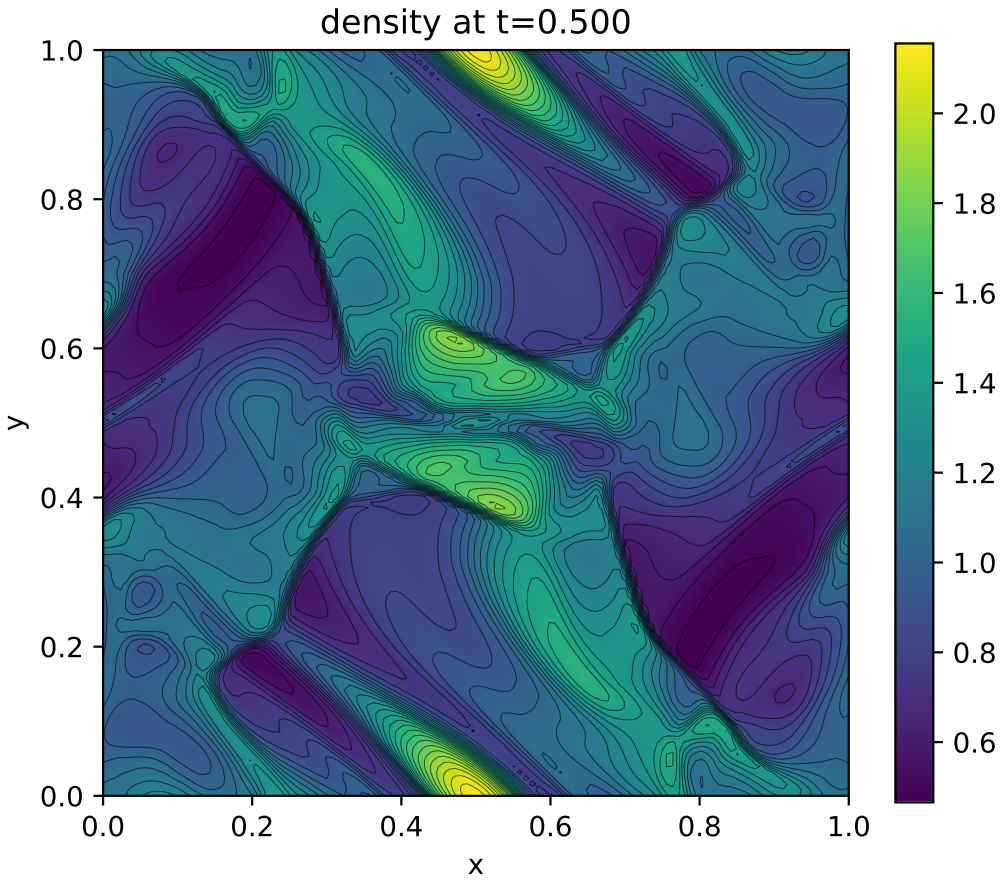
Wave regions across the x-face.

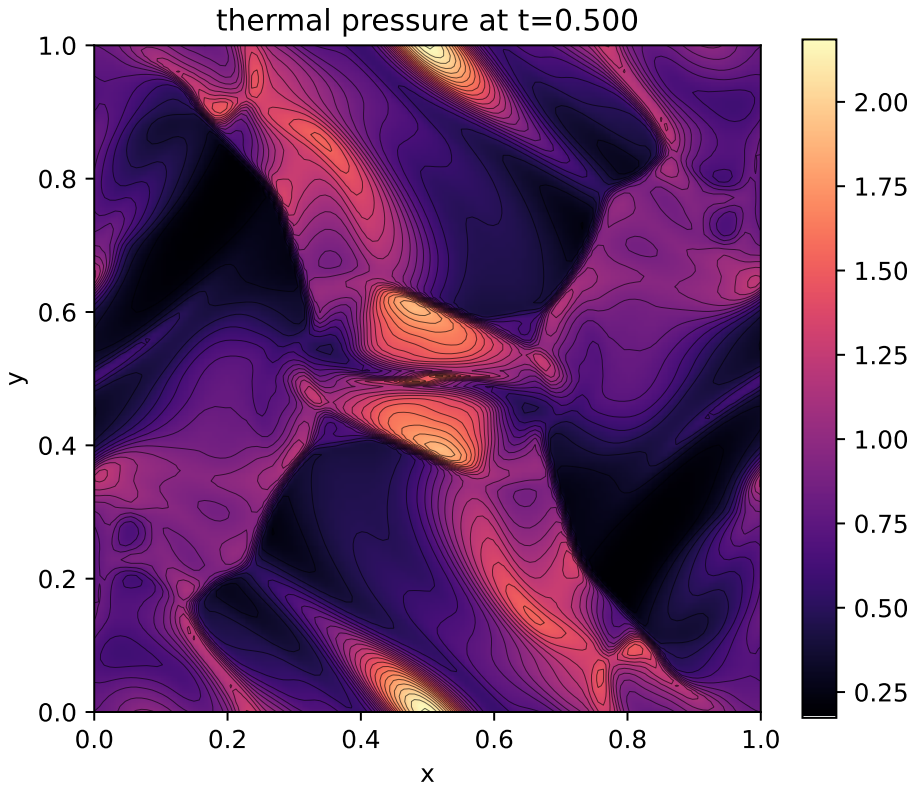
Density at t = 0.5, 128^2

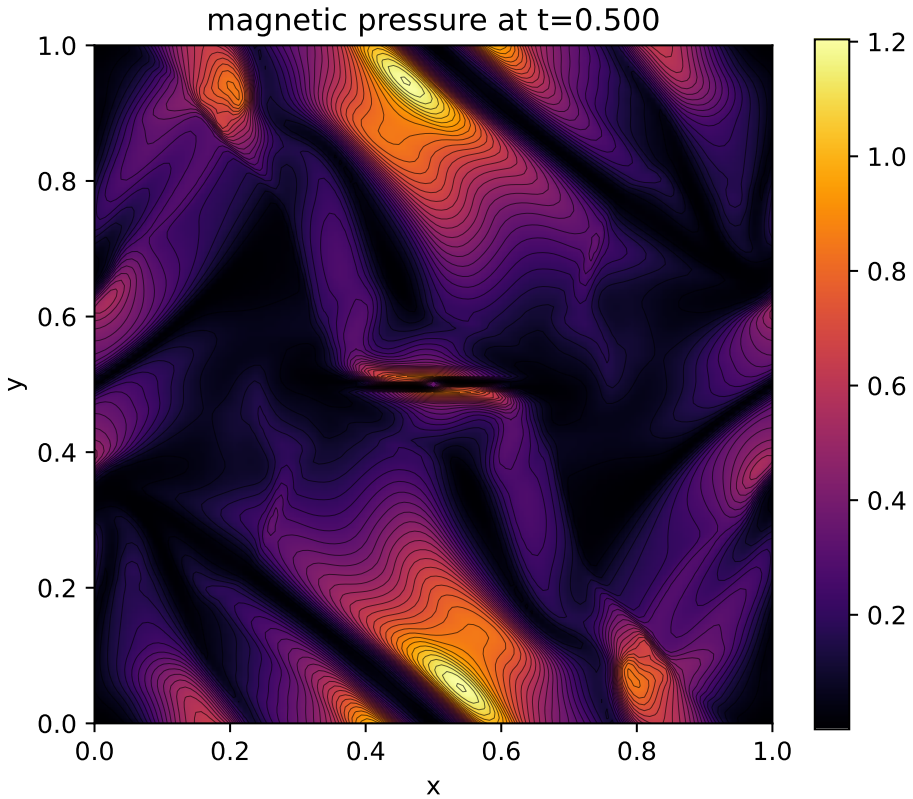


$\rho(x, y,\ t = 0.5)$

- Central bow-tie current sheet.
- Four symmetric oblique shocks bounding low-density bays.
- Qualitative match with [3, 4] at the same time slice.

Thermal and magnetic pressure, t = 0.5



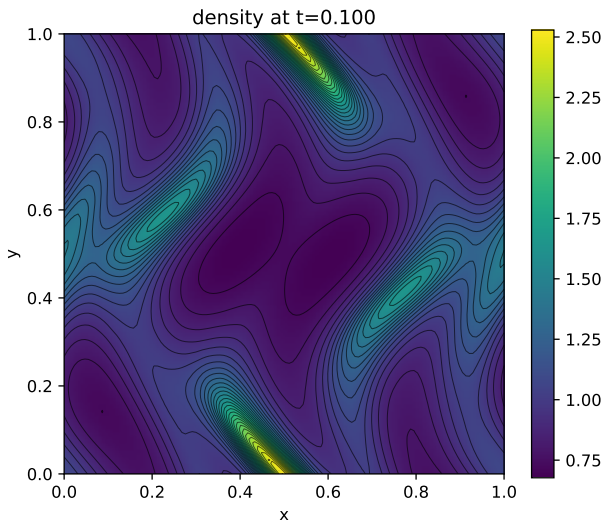


$$p = (\gamma - 1)\left(E - \tfrac{1}{2}\rho|\mathbf{v}|^2 - \tfrac{1}{2}|\mathbf{B}|^2\right)$$

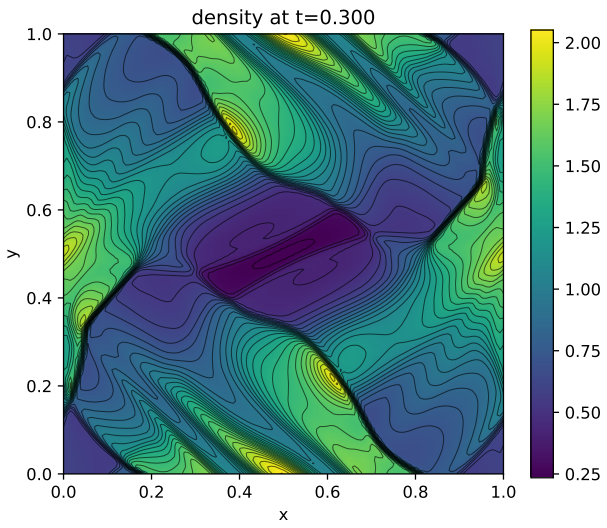
$$p_{\text{mag}} = \tfrac{1}{2}|\mathbf{B}|^2$$

- Thermal pressure tracks density with extra shock heating near the current sheet.
- Magnetic pressure peaks at flux pile-up in the four current-sheet arms.

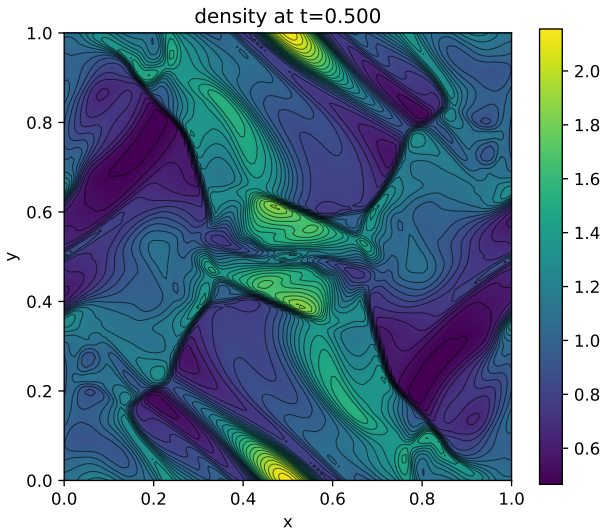
Formation of the current sheet



$\rho(x, y,\ t = 0.1)$



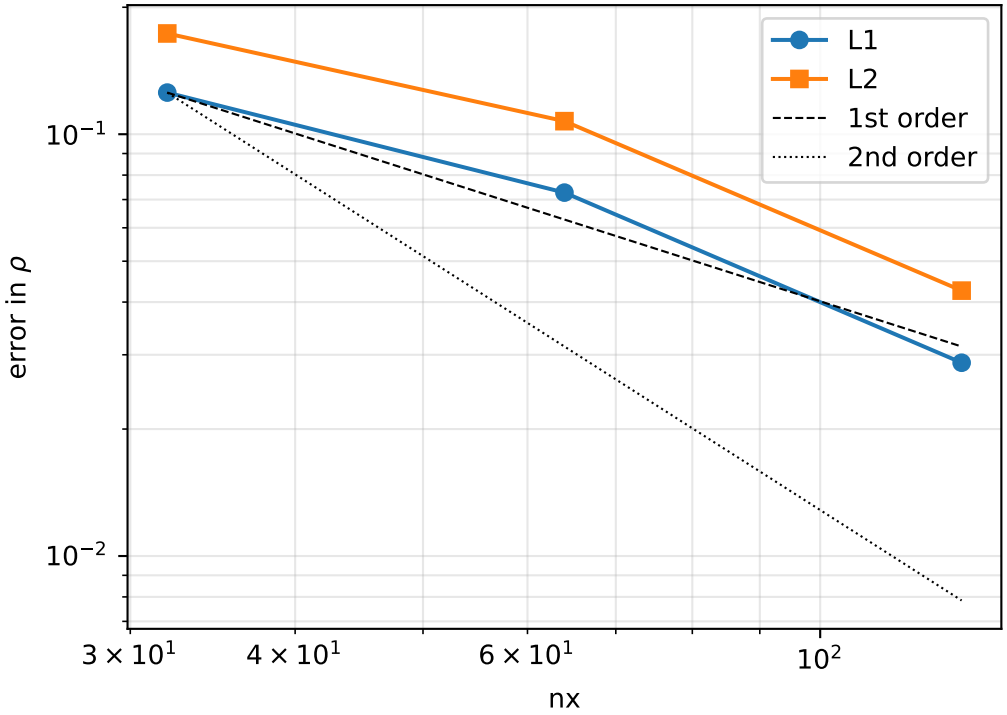
$\rho(x, y,\ t = 0.3)$



$\rho(x, y,\ t = 0.5)$

- $t = 0.1$: still smooth, the vortices are just forming.
- $t = 0.3$: diagonal shocks have formed; bow-tie compression starts.
- $t = 0.5$: fully-developed current sheet and bounding shocks.

Convergence in density

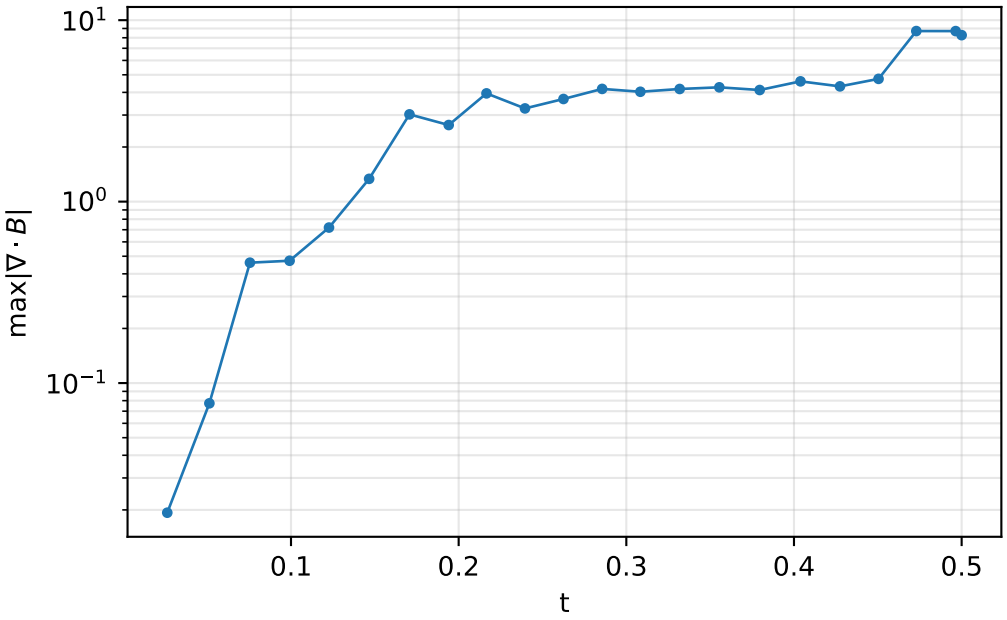


$\|\rho_h - \rho_{\text{ref}}\|_p$ vs grid resolution, ρ_{ref} from 256^2

nx	L^1	L^2
32	1.26e-1	1.73e-1
64	7.27e-2	1.07e-1
128	2.87e-2	4.26e-2

- $32 \to 64$: rate ≈ 0.79.
- $64 \to 128$: rate ≈ 1.33.
- Between 1st and 2nd order: expected for minmod on a shock-dominated flow [12].

Divergence control and conservation



$\max_{i,j} |\nabla \cdot B|_{i,j}(t)$, centered differences on cell-centered B

- GLM [9] bounds $\max|\nabla \cdot B|$ once shocks form.
- Centered-difference $|\nabla \cdot B|$ is pessimistic; GLM-natural measure is smaller.
- Mass, momentum, energy conserved to machine precision ($\sim 10^{-16}$ per step, see `run.log`).

Accuracy vs limiter choice

- At 128^2 here: $\rho_{\text{min}} \approx 0.47$, $\rho_{\text{max}} \approx 2.16$.
- Athena++ with MC at the same resolution: $\rho_{\text{min}} \sim 0.1$, $\rho_{\text{max}} \sim 3.5$ [4].
- Gap is the limiter: minmod is the most diffusive TVD choice and smears current sheets [12].
- MC or van Leer would close it—but cost robustness in low-beta cells.

Why between first and second order

- MUSCL is second-order in smooth regions [12].
- TVD limiters drop to first order at extrema and at shocks.
- Orszag-Tang at $t = 0.5$ is shock-dominated $\Rightarrow$ global rate between 1 and 2.
- Observed: ≈ 0.8 coarse (shocks dominate), ≈ 1.3 fine (smooth regions start to dominate).

Limitations

- Vectorized NumPy: 128^2 in $\sim 7\,\mathrm{s}$; 256^2 is the practical ceiling on a laptop.
- No constrained-transport option—GLM is simpler but less strict on $\nabla \cdot B$ [8].
- 2D Cartesian; no AMR.
- Linear reconstruction (MUSCL); no WENO, no higher-order time integration.

Conclusions

- 2D ideal MHD solver from scratch: Godunov FVM, MUSCL-minmod, HLLD [7], GLM [9], SSP-RK2 [13].
- Orszag-Tang at 128^2 reproduces the canonical $t = 0.5$ density/pressure pattern [3, 4].
- Mass, momentum, energy conserved to machine precision; GLM bounds $\max|\nabla \cdot B|$.
- Convergence rate between 1 and 2, as expected for MUSCL-minmod on shock-dominated flow.

Natural extensions: constrained transport, resistive MHD, 3D, AMR, Numba/JAX.

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